



Project Insights & Analytics Report: Shoppers Spectrum

An End-to-End Retail Intelligence & Cloud-Deployed Dashboard Engine

1. Executive Summary

The **Shoppers Spectrum Dashboard** is an enterprise-ready retail analytics platform designed to bridge the gap between raw e-commerce transaction tracking and automated, data-driven revenue operations. Built completely in Python and deployed as an interactive web service via Streamlit, the system implements a dual-engine architecture:

- **Track A: Product Recommendation Engine** leveraging Item-Based Collaborative Filtering to automate cross-selling and maximize basket sizes.
 - **Track B: Customer Segmentation Engine** combining Recency, Frequency, and Monetary (RFM) analytics with unsupervised K-Means Clustering to group accounts into high-yield marketing profiles.
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2. Core Data Engineering Pipeline

The project ingests raw transaction ledger records ([online_retail.csv](#)), evaluating initial fields like [InvoiceNo](#), [StockCode](#), [Description](#), [Quantity](#), [InvoiceDate](#), [UnitPrice](#), [CustomerID](#), and [Country](#). To ensure absolute data integrity for downstream machine learning algorithms, a multi-stage cleanup pipeline was structured in the [Shoppers Spectrum Notebook](#):

- **Deduplication:** Identified and stripped **5,268 explicit duplicate rows** from the transactional matrix.
- **Customer Isolation:** Filtered out all anonymous records lacking a unique account identifier ([CustomerID.isna\(\)](#)).
- **Returns & Cancellations Scrubbing:** Dropped order records with negative quantities and items where the [InvoiceNo](#) started with "C", isolating pure purchasing history.
- **Zero-Price Correction:** Extracted and purged administrative or bad rows where the item [UnitPrice](#) ≤ 0 .
- **Feature Enrichment:** Formulated a net transactional feature $\text{TotalAmount} = \text{Quantity} * \text{UnitPrice}$. Extracted temporal metrics ([MonthYear](#) profiles and order checkout [Hour](#) integers) to monitor behavioral patterns.

The cleaned dataset stabilized into a dense core of **392,692 reliable purchase vectors** saved out to the root directory as [cleaned.csv](#).

3. Exploratory Data Analysis (EDA) & Visual Insights

Before routing the cleaned dataset into machine learning environments, a comprehensive behavioral analysis was executed across temporal, geographical, and monetary dimensions.

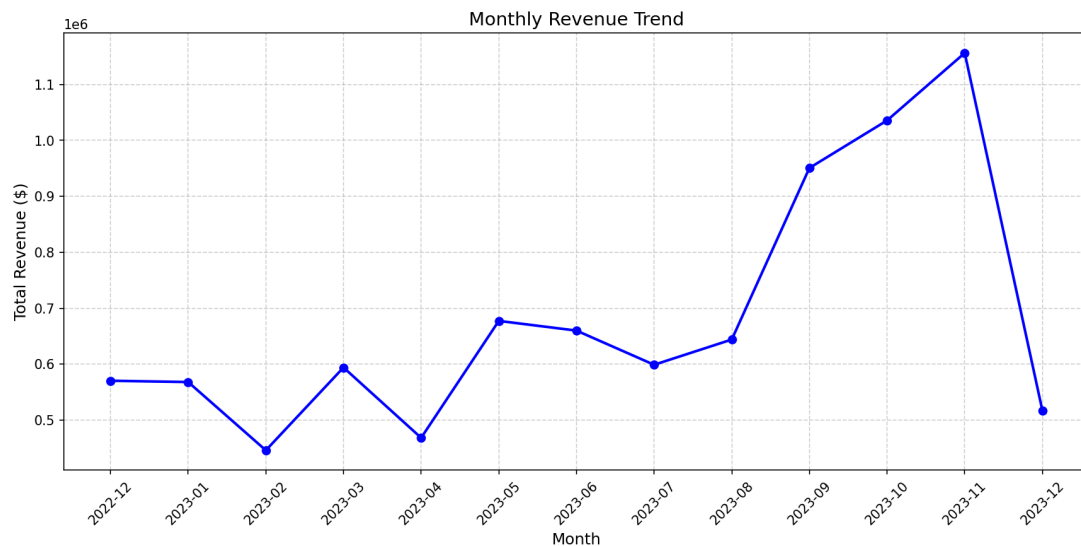
A. Temporal & Operational Trends

To optimize retail operations and predict server load capacities, transactional data was plotted across historical and hourly cycles.



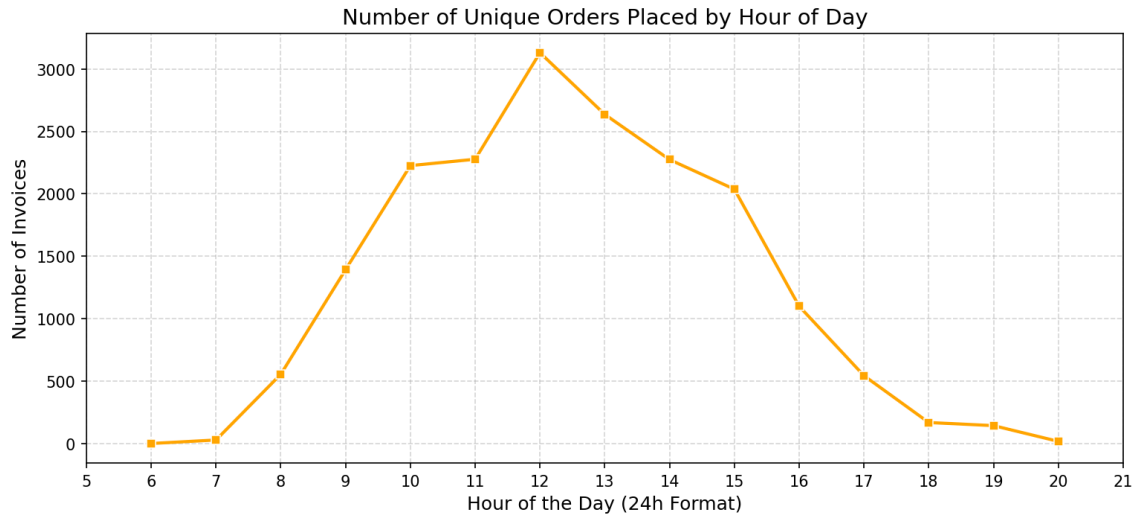
Monthly Sales Trend

- **Analysis:** Monitors macro-level transactional shifts, mapping seasonal peaks, promotional quarters, and baseline growth periods across the e-commerce roadmap.



Customer Hourly Purchase Pattern

- **Analysis:** Isolates high-volume traffic intervals during the day. This data tells marketing exactly when to trigger targeted push notifications and helps engineering schedule backend system maintenance during low-activity intervals.

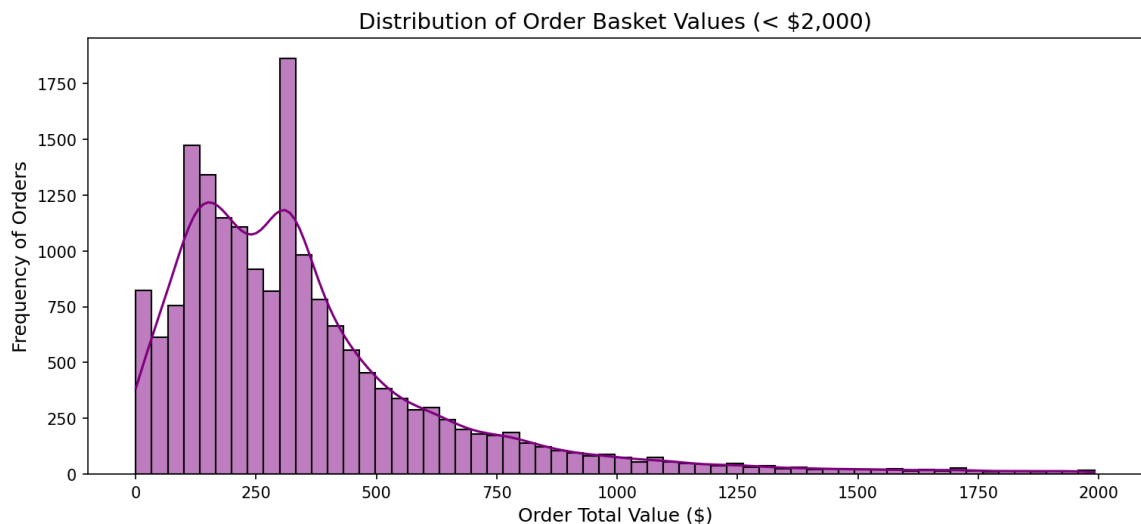


B. Revenue & Core Performance Indicators

These visualizations highlight the core financial drivers of the retail system by breaking down key revenue nodes.

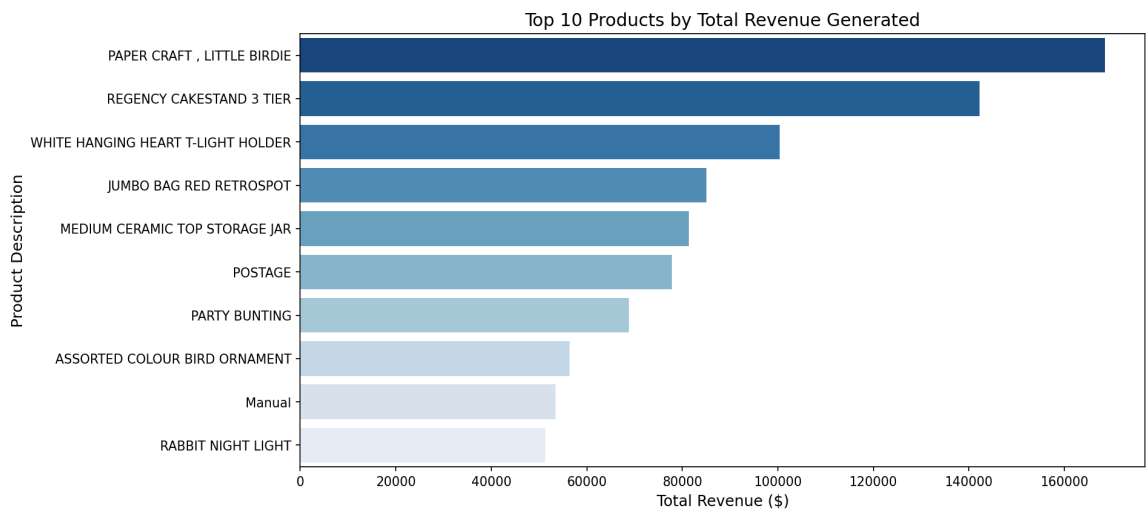
Revenue Analysis

- **Analysis:** Provides a high-level breakdown of profit centers, tracking how revenue scales alongside operational changes and identifying standard basket trends.



Top-Selling Products (by Revenue)

- **Analysis:** Ranks inventory items by total revenue generated. This directly informs stock reordering schedules and highlights which core items should feed into the **Product Recommendation Module**.

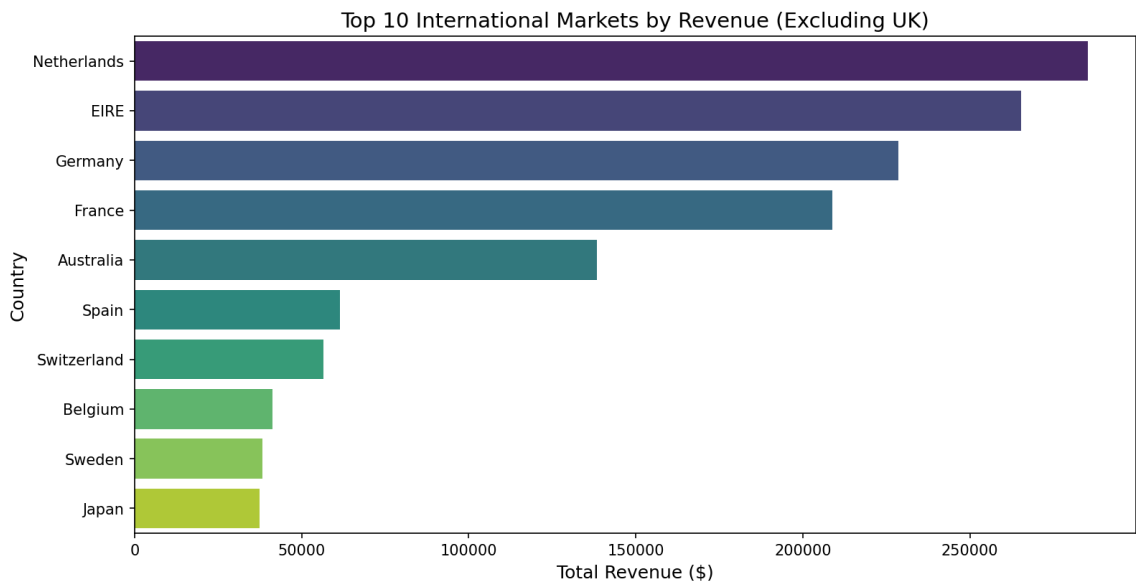


C. Client Profiles & Demographics

Understanding *who* buys and *where* they check out forms the bedrock of the downstream target models.

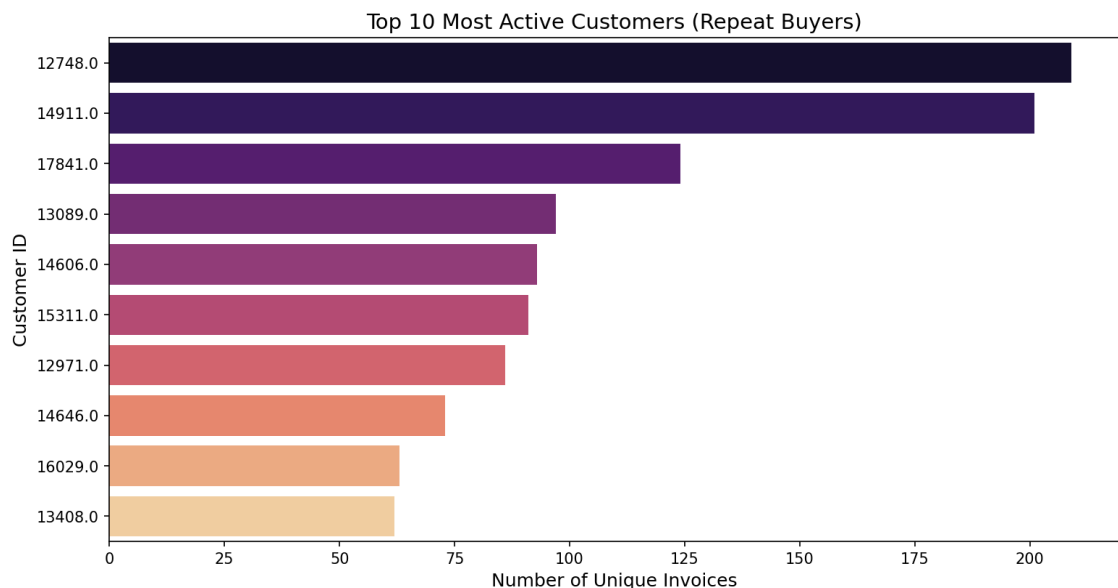
🌐 Country-wise Sales

- **Analysis:** Visualizes international demand to pinpoint geographical strongholds. This helps optimize localized distribution centers and refine region-specific marketing budgets.



👥 Most Active Customers

- **Analysis:** Spotlights top repeat buyers based on purchase counts. This provides early validation for the **High-Value Customer** indicators used in our segmentation model.

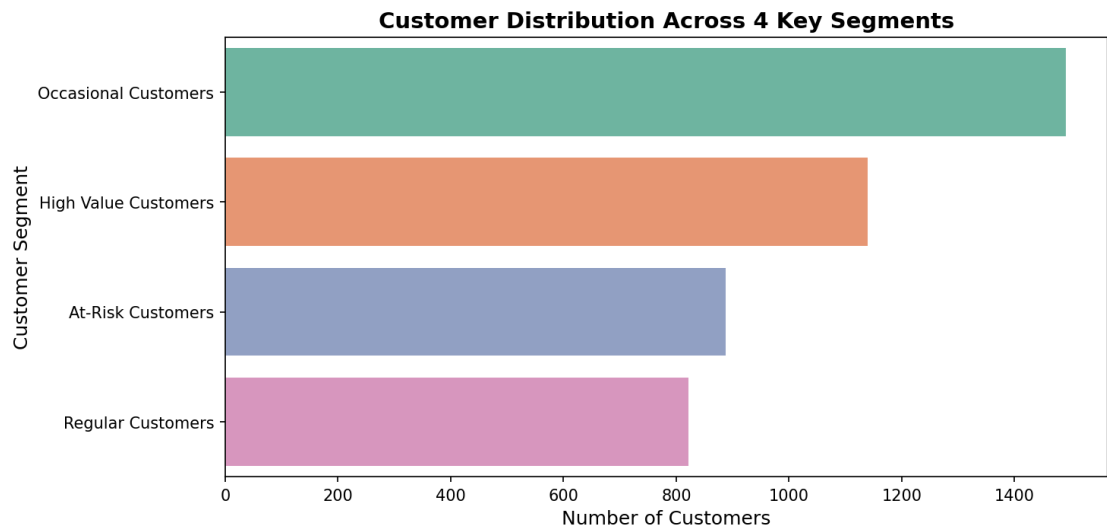


D. Model Target Validation

This visualization serves as the mathematical validation step connecting raw data exploration to unsupervised machine learning.

🎯 Customer Distribution Across 4 Key Segments

- **Analysis:** Shows the volume distribution across our four target profiles (High-Value, Regular, Occasional, and At-Risk). This ensures the business has a well-balanced distribution before executing personalized outreach campaigns.



4. Machine Learning Architectures

Track A: Product Recommendation (Collaborative Filtering)

Rather than relying on basic user tracking, this engine maps product relationship indexes.

1. **Interaction Matrix Construction:** Transactions are pivoted into a lean item-to-user matrix, where each item's footprint is defined by its customer purchasing history.
2. **Cosine Similarity Formulation:** Pairwise similarity scores are calculated across all unique products. If two products are consistently checked out by the same groups of customers, their cosine vector distance shrinks toward 1.
3. **In-Memory Lookups:** The finalized similarity matrix is cached as a DataFrame binary (`item_similarity.pkl`). When a merchant selects a specific item on the dashboard, the application pulls the top corresponding item matches instantly.

Track B: RFM Customer Clustering (K-Means)

To group client accounts without relying on arbitrary human intuition, unsupervised machine learning was deployed on aggregated behavioral dimensions:

- **Recency (\$R\$):** The exact number of days elapsed between a customer's latest checkout date and the dataset tracking cutoff point.
- **Frequency (\$F\$):** Cumulative count of unique invoices generated under a specific customer account.
- **Monetary (\$M\$):** Net historical financial spend contributed by the client account.

Skewness Correction & Normalization

Initial checks revealed severe right-skewness across the dimensions, particularly for Frequency (skewness of 12.06) and Monetary metrics (skewness of 19.33). This skew can heavily distort distance-based clustering algorithms like K-Means.

To correct this, the mathematical pipeline applies:

1. An `np.log1p` log transformation across all raw rows, shrinking outliers and pulling the records into a normal distribution curve.
2. A `StandardScaler` normalization pass, centering the data around a mean of 0 with a standard variance of 1 so no single dimension disproportionately scales the cluster results.

Model Selection (Elbow & Silhouette)

By analyzing the drop in *Inertia* (Within-Cluster Sum of Squares) alongside maximizing the *Silhouette Score* across a testing range of $K = [2, 10]$, an optimal cluster configuration of **K = 4 distinct clusters** was locked in. The trained model architecture is serialized directly to `Models/kmeans_model.pkl`.

5. Target Segment Profile Specifications

The final machine learning pipeline segments the client database into four distinct performance bands:

Profile Segment	Behavioral Profile (RFM Mapping Specs)	Tactical Action Plan & Marketing Strategy
 High-Value Customer	Outstandingly low recency values (recently bought), combined with top-tier volume frequencies and massive lifetime monetary spend.	Enroll in VIP white-glove rewards programs, grant early access to premium product lines, and leverage point multipliers.
 Regular Customer	Steady interaction timelines, consistent return histories, and stable average order baskets.	Deploy automated product bundling packages and cross-sell subscription upsells to maximize customer lifetime value (LTV).
 Occasional Customer	Infrequent order interactions, flat buying spikes, and moderate or low financial margins.	Trigger flash discounts, target with seasonal email re-engagement push notifications, and offer low-threshold free shipping.
 Customer At-Risk	Stagnant account timelines with severe recency gaps, dropping performance numbers, and low transaction histories.	Launch aggressive win-back promotional campaigns, offer direct account manager check-ins, or provide feedback surveys.

6. Web Application & Cloud Architecture

The user interface is built as a responsive cloud dashboard written entirely inside [app.py](#).

Layout & Optimization Design

- **Wide Framework Integration:** Utilizes `st.set_page_config(layout="wide")` to render independent data visualization grids side-by-side, preventing vertical layout clutter.
- **Clean Session State Management:** Implements customized `st.session_state` keys attached to action buttons. This allows the user to trigger dynamic "Clear Selection" resets without reloading the entire application page.
- **Theme-Independent High Contrast Override:** Employs customized inline HTML containers and CSS overrides. This guarantees that all text components remain perfectly legible across both dark and light UI themes, completely eliminating faded text bugs.